

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 8 Summer 2025 (Regular)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTMEC801F - NON-CONVENTIONAL ENERGY RESOURCES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1 An artificial pacemaker of tens of microwatt is made up of_____.

1. Li-ion battery
2. Secondary battery
3. Primary battery
4. All of the Above

2 The region of the atmosphere where human activity has the most impact is.

1. Troposphere
2. Stratosphere
3. Mesosphere
4. Ionosphere

3 A normal human being cannot see the things in the complete darkness because eyes see the objects only through_____.

1. Reflected visible light
2. Reflected IR
3. Inherent IR of the object being seen
4. Reflected UV

4 Flat plate collectors are made up of_____.

1. Metals
2. Metals with high thermal conductivity
3. Metals which can absorb heat easily and transfer the heat through conduction
4. All of the above

5 When a p-n junction is at equilibrium, its _____ is flat.

1. Band structure
2. Valence band at the junction
3. Conduction band at the junction
4. Fermi energy

6 Solar cell is a _____ while battery is a _____ based device.

1. Voltage source, Current source
2. Current source, voltage source
3. Voltage source, electron source
4. Ionic source, electronic source

7 Windmill's energy generation depends on the _____.

1. Volume of air going through it
2. Cube power of wind speed
3. Height of the windmill
4. All of the Above

8 What are the challenges with OTEC plant?

1. Pressure
2. Depth
3. Water Current
4. All of the Above

9 What kind of material/coatings is used for coating turbine blades?

1. Metallic coating
2. Composite
3. Ceramic
4. None

10 Battery and fuel cell _____.

1. Both are electrochemical devices
2. Both can be used for electric vehicles
3. Both a & b
4. None

11 Electrolyte used in fuel cell and Li ion battery are ____.

1. Same
2. Different
3. Electronic conductors
4. None

12 Which of the following are energy conversion devices

1. Battery
2. Fuel Cell
3. Capacitor
4. None

Q2. Solve the following.

- A) Write a short note on scale of quantities
- B) Discuss energy consumption by different sector

6
6

Q3. Solve the following.

- A) Write a short note on solar spectrum
- B) Enlist solar thermal devices & Explain solar flat plate collector

6
6

Q4. Solve Any Two of the following.

- A) Discuss performance & durability of solar photovoltaic device

6

- B) Explain with neat diagram Horizontal axis wind turbine (HAWT). 6
- C) Write a note on biomass. 6
- Q5. Solve Any Two of the following.
- A) Explain in detail geothermal energy with schematic diagram. 6
- B) Give detailed classification of battery & discuss Li-Ion Battery in detail. 6
- C) Write a short note on parameters which will affect performance of batteries 6
- Q6. Solve Any Two of the following.
- A) Give Types of Fuel cell & explain in detail alkali fuel Cell (AFC) 6
- B) Write a short note on fly wheel. 6
- C) What is Supercapacitor? Discuss in detail construction & Working of super capacitor 6

*** End ***